

## FREQUENCY RESPONSE OF AN AC INVERTING AMPLIFIER

### EXPERIMENT NUMBER 02

#### OBJECTIVES:

- Describe the importance of feedback and frequency response.
- Design an AC inverting amplifier with a desired voltage gain.
- Select proper values of external components for a desired bandwidth.
- Discuss the differences between an AC amplifier with a dual supply and one with a single supply voltage.

#### APPARATUS:

DC dual power supply, multimeter, signal generator, oscilloscope, breadboard, 741 op-amp, 470 k $\Omega$ , 2  $\times$  100 k $\Omega$  and 3  $\times$  10 k $\Omega$  resistors, 0.1  $\mu$ F, 0.01  $\mu$ F capacitors and connecting wires.

#### THEORY AND DIAGRAMS:

##### *Frequency Response and Bandwidth*

The bandwidth of an ideal operational amplifier is infinite. In other words, it is assumed that an ideal operational amplifier can amplify any frequency signal. Furthermore, it is also assumed that an ideal operational amplifier has an infinite gain, irrespective of the frequency of the input signal. However, real operational amplifiers do not have infinite gain or bandwidth but are characterised by “Open Loop Gain”, which is defined as the gain of the amplifier without any external feedback signals connected to it. The curve, which represents the variation of the voltage gain with frequency, is known as the Frequency Response Curve. The frequency response curve for a typical operational amplifier is illustrated in Figure 1. As you can see, the open loop gain is about 100 dB at 0 Hz.

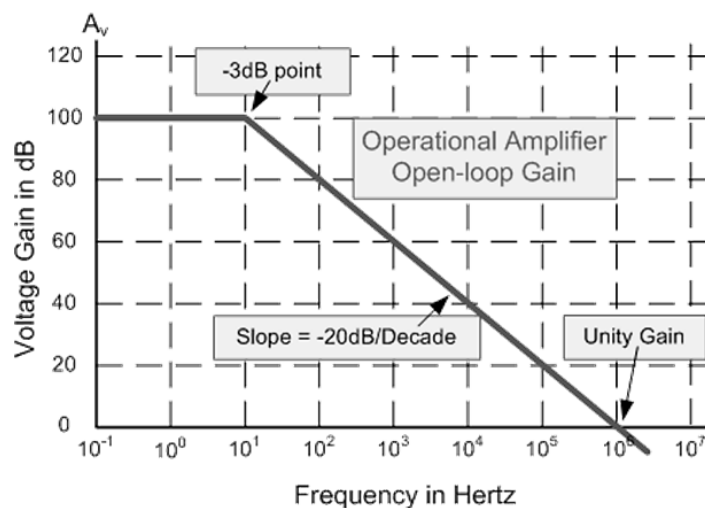


Figure 1

It is clear that the bandwidth is very small (about 10 Hz) if the op-amp is used without any feedback. If required, the bandwidth can be extended by employing a feedback loop to the op-amp. This is possible because it can be shown that the Gain Bandwidth Product (or GBP) remains constant. Mathematically this can be expressed as,

$$GBP = Gf_B = f_T$$

where,  $G$  = Low frequency gain

$f_B$  = 3 dB Bandwidth (i.e. the frequency range over which the voltage gain of the amplifier is above 70.7% or -3 dB of its maximum output)

$f_T$  = Unity gain frequency.

Therefore, if an op-amp has a unity gain frequency,  $f_T$  of 1 MHz, it can be used to make a feedback amplifier with a gain of one and a bandwidth of 1 MHz or with a gain of 10 and a bandwidth of 100 kHz, etc.

### ***AC Inverting Amplifier***

Inverting and noninverting amplifiers can respond to AC input signals as well. However, suppose we want the AC response characteristics of the amplifier. In that case, we have to use an AC amplifier with a coupling capacitor when AC input is superimposed on some AC level. For example, an audio system consisting of several stages, a DC level can be produced due to thermal drift, component tolerance, and variations. To prevent the application of such DC levels, the coupling capacitors must be used between the signals.

The coupling capacitor,  $C_s$  blocks the DC voltage and also set the low-frequency cut-off limit,  $f_L$  which is given by,

$$f_L = \frac{1}{2\pi C_s (R_{if} + R_s)}$$

where,  $R_{if}$  = AC input resistance of the gain stage

$R_s$  = source resistance or ac output resistance of the dividing stage

Consider the AC inverting amplifier circuit shown in Figure 2 below.

If the input to the circuit is a harmonic signal of frequency  $\omega$ , it can be shown that the output signal voltage,  $V_o$  is given by,

$$V_o = -\frac{j\omega C_i R_f}{1 + j\omega C_i R_i} V_{in}$$

where,  $j^2 = -1$

For sufficiently large values of  $\omega$ , (larger than a lower cutoff value determines by  $C_i$  and  $R_i$ ), the above equation reduces to,

$$V_o = -\frac{R_f}{R_i} V_{in}$$

and the circuit acts as an inverting amplifier.

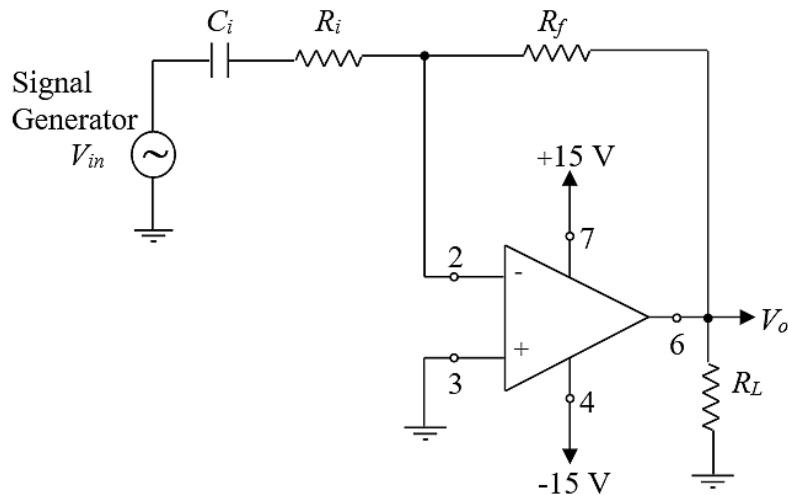


Figure 2

#### PROCEDURE:

1. Connect the circuit as shown in Figure 3.

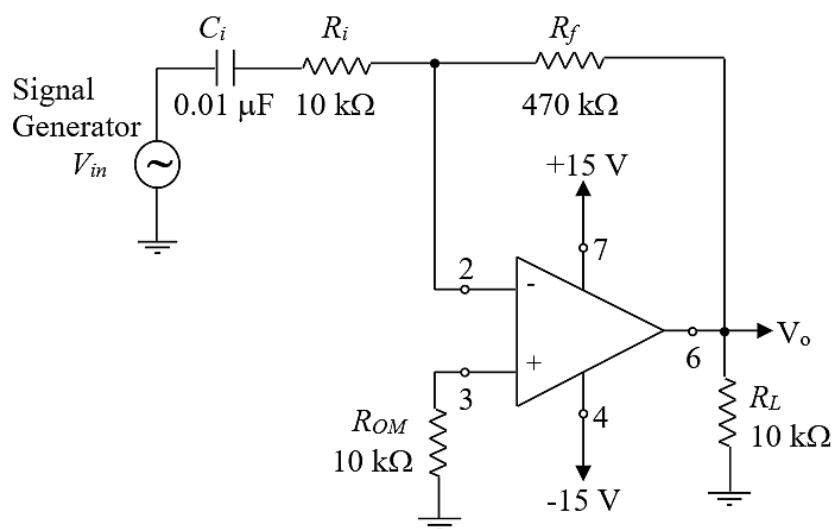


Figure 3

2. Set  $V_{in} = 100$  mV peak-to-peak sine wave at 100 Hz. Measure the input and output voltages.
3. Repeat step 2 for the frequencies given in the sample data sheet. If needed, obtain more data.
4. Next, connect the circuit shown in Figure 4 and repeat steps 2 and 3.

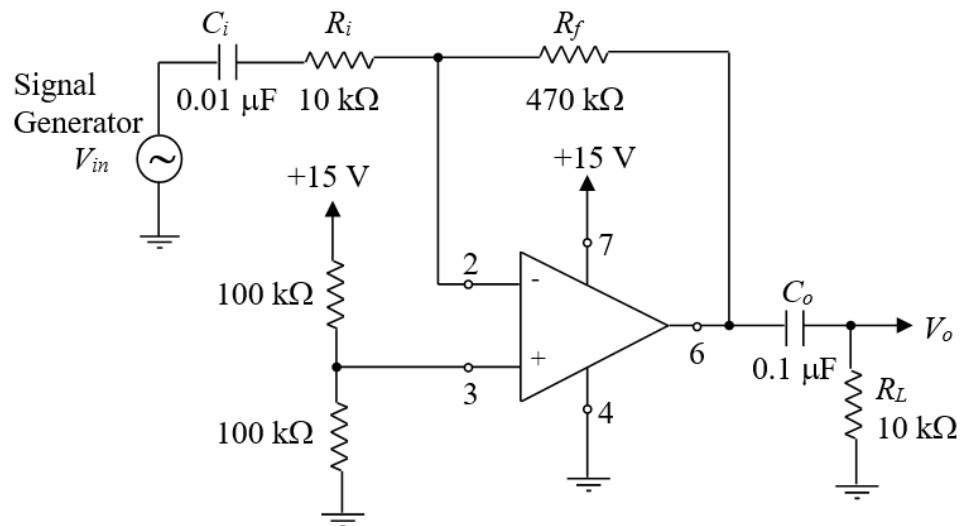


Figure 4

- Complete the columns given in the sample data sheet shown in Table 1.
- Plot the frequency versus gain (in dB) curves for the inverting amplifier with dual and single supply on semi-log graph papers.
- From the graphs, find the low-cutoff and high-cutoff frequencies and hence determine the bandwidths.

Table 1: Sample Data Sheet

| Input voltage<br>$V_{in}$ pp (V) | Input frequency<br>(Hz) | Output voltage<br>$V_o$ pp (V) |               | Voltage gain<br>$A_v$ |               | Voltage gain<br>(dB) |               |
|----------------------------------|-------------------------|--------------------------------|---------------|-----------------------|---------------|----------------------|---------------|
|                                  |                         | Dual supply                    | Single supply | Dual supply           | Single supply | Dual supply          | Single supply |
| 100 mV                           | 100                     |                                |               |                       |               |                      |               |
|                                  | 200                     |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |
|                                  | 1,000                   |                                |               |                       |               |                      |               |
|                                  | 2,000                   |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |
|                                  | 10,000                  |                                |               |                       |               |                      |               |
|                                  | 20,000                  |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |
|                                  | 100,000                 |                                |               |                       |               |                      |               |
|                                  | 200,000                 |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |
|                                  | .                       |                                |               |                       |               |                      |               |

**DISCUSSION:**

1. Calculate the theoretical bandwidths for the circuits. How are the calculated bandwidths compared with the measured bandwidths?
2. In the circuit shown in Figure 3, if  $C_i$  is changed to 0.1  $\mu\text{F}$ , what would be the new low-cutoff frequency?
3. What determines the high-cutoff frequency of an amplifier?
4. For a high-cutoff frequency of 50 kHz, what should be the gain of an amplifier?
5. What are the advantages and disadvantages of a single-supply amplifier?